HARDWARE × MODELS × REAL-WORLD SPEED

Intelligence. On your desk.

A field guide to capable local coding and tool-using agents.
14 hardware configurations. Seven local models + four Claude Opus reference points.

Model capability, memory, speed and cost.
Reported benchmarks and clearly labelled estimates for local AI hardware.

01 / The hardware-to-model map

M reported measurement E estimate

35 tok/s gate: choose the highest-scoring fitting model whose decode range reaches ≥35 tok/s, then sort by score and speed ↑. One active request; a qualifying upper bound is not a guaranteed minimum.

DEVICE / CONFIGURATION	APPROX. USD	MEMORY INSTALLED	BANDWIDTH GB/S	AI COMPUTE DENSE PEAK	BEST-FIT SHORTLIST MODEL / PRECISION	DECODE TOKENS / SEC
MacBook Pro · M5 [H1] 10-core GPU	**$2.5k** System	**32 GB** Unified	153	**~30 TOPS** INT8 GPU · E	Qwen3.6-35B-A3B · 4-bit MLX oMLX 0.3.6 · ~20 GB peak	**51–60** M 1K–4K · two runs · B12
MacBook Pro · M5 Pro [H1] 20-core GPU	**$3.7k** System	**64 GB** Unified	307	**~67 TOPS** INT8 GPU · E	Qwen3.8-27B · dynamic Q4 MTPLX · native MTP drafting	**21–44** E Upper end: repetitive editing
GeForce RTX 4090 [H3]	**$2–2.5k** Card est.	**24 GB**	1,008	**661 TFLOPS** FP8	Qwen3.8-27B · Q4_K_S llama.cpp · no MTP	**38–46** M 64K → 4K · B10
Radeon AI PRO R9700 [H4]	**$1.7k** Card	**32 GB**	640	**383 TFLOPS** FP8	Qwen3.8-27B · Q4-class + MTP llama.cpp Vulkan · MTP=2	**51.8** M AMD vendor · preliminary · B8
GeForce RTX 5090 [H5]	**$6.4–7.4k** Card est.	**32 GB**	1,792	**1,676 TFLOPS** FP4	Qwen3.8-27B · Q4_K_M LM Studio / CUDA	**59** M 6.1K · no MTP · B3
Ryzen AI Max+ 395 / Halo [H2] Radeon 8060S	**$2–4k** System est.	**128 GB** GPU share varies	256 LPDDR5X-8000	**110 T ops/s** Mixed peak: GPU + NPU 60 FP16 + 50 INT8	Qwen3.8-Flash-Next · UD-IQ4_XS llama.cpp ROCm · MTP + n-gram drafting	**32–47** M 8K new code → rewrite · B7
DGX Spark [H7] Single NVIDIA GB10 system	**$4.7k** System	**128 GB**	273	**500 TFLOPS** FP4	Qwen3.8-Flash-Next · NVFP4 vLLM + MTP3 · table on SSD	**37–46** M Prose → code · B5
MacBook Pro · M5 Max [H1] 40-core GPU	**$6k** System est.	**128 GB** Unified	614	**~130 TOPS** INT8 GPU · E	Qwen3.8-Flash-Next · dynamic Q4 MTPLX 2.12 + MTP · n-gram table on SSD	**63.5** M 65.5K prompt · B1
RTX PRO 6000 Blackwell [H6] Workstation Edition	**$16–20k** Card	**96 GB**	1,792	**2,015 TFLOPS** FP4	Qwen3.8-Flash-Next · NVFP4 vLLM · n-gram table in ~100 GB host RAM	**81–88** M No MTP; 120–173 MTP · B9
Mac Studio · M5 Ultra [H8] 80-core GPU	**$12k** System est.	**256 GB**	1,200	**~260 TOPS** INT8 GPU · E	GLM-5.3-Flash · oQ4e / mixed 4/8-bit oMLX + DFlash2 drafter	**44–62** M 4K → short prompt · B17
2 × DGX Spark [H7]	**$9.4k** Systems	**256 GB** 128 GB per node	2 × 273 Link: 25 GB/s 200 Gb/s RoCE	**1,000 TFLOPS** FP4 · 2 devices	GLM-5.3-Flash · EXL3 4-bit TP2 · DFlash2 drafter · 200G RoCE	**47–63** M Prose → code · B16
2 × RTX PRO 6000 Blackwell [H6]	**$32–40k** Cards est.	**192 GB** 96 GB per GPU	2 × 1,792 Link: ≤63 GB/s PCIe 5.0 ×16*	**4,030 TFLOPS** FP4 · 2 GPUs	GLM-5.3-Flash · IQ4_XS Layer split · ~157 GB weights · fit projected	**41–70** E Best ~55 · B13 anchor
M5 Ultra · 512 GB [H8] Ships late October	**$15–18k** System est.	**512 GB**	1,200	**~260 TOPS** INT8 GPU · E	GLM-5.3 · IQ4_XS / oQ4 Needs DFlash2 drafter · ~365–420 GB weights	**18–40** E Drafting · 15–25 without
DGX Station · GB300 [H9] OEM order · deskside	**$99–189k** System	**748 GB** 252 HBM + 496	7,100 / 396 HBM / LPDDR5X	**15,000 TFLOPS** FP4	GLM-5.3 · IQ4_XS, split HBM / CPU Needs DFlash2 drafter + split-memory backend	**19–40** E Drafting · 16–25 without

Compute: Dense peaks only; no sparsity multiplier. Macs show the higher INT8 GPU peak; other GPUs show TFLOPS at the stated precision. Macs exclude the separately implemented Neural Engine: its dense rating is unverified; GPU FP16 and INT8 peaks cannot be added. Halo sums 60 FP16 TFLOPS (GPU) + 50 INT8 TOPS (NPU): a mixed-precision theoretical aggregate for separate concurrent engines, not measured LLM throughput. Shared power/memory can limit simultaneous peaks. **Links:** per-direction ceilings, separate from memory bandwidth; *dual-PRO assumes full ×16 slots, with actual transfers depending on host topology. **Prices:** approximate USD; cards exclude host. **M / E:** third-party measurement / forecast. 128K memory budgets exclude unquantified DFlash additions. Two PRO Server cards separately report **164.8 tok/s** with another quantization (B19).

02 / Local models & the frontier trajectory

BASELINE Claude Opus · cloud yardstick, not runnable locally AA Intelligence Index v4.3.2 · higher is better

10 independent evaluations: agents 30% · coding 20% · scientific reasoning 20% · general 30%. **Opus trajectory:** 4.6 (Feb) 32* → 4.8 (May) 42 → 5 (Jul) 51 → 5.5 (Sep) 58. The best local-fitting model, GLM-5.3 (45), sits between May's Opus 4.8 and July's Opus 5.

MODEL / TESTED SETTING	AA SCORE	PARAMETERS / TYPE	MEMORY ESTIMATE · 128K CONTEXT · 1 ACTIVE REQUEST
Claude Opus 5.5 BASELINE Max · cloud ceiling Released 22 Sep 2026	58	**Undisclosed** Architecture undisclosed	**Cloud only · weights and serving memory undisclosed**
Claude Opus 5 BASELINE Max · cloud reference Released 24 Jul 2026	51	**Undisclosed** Architecture undisclosed	**Cloud only · frontier trajectory reference**
GLM-5.3 Max Released 18 Aug 2026 · API notes	45	**~753B total · ~40B active** MoE	**365 GB weights + ~13 GB cache ≈ 378 GB resident** IQ4_XS · compressed MLA, 16-bit cache · SSD tables: none · runtime: +8–16 GB
Claude Opus 4.8 BASELINE Max · cloud reference Released 28 May 2026	42	**Undisclosed** Architecture undisclosed	**Cloud only · frontier trajectory reference**
GLM-5.3-Flash Reasoning Released 26 Aug 2026	42	**320B total · 18B active** MoE	**157–200 GB weights + ~2 GB cache ≈ 159–202 GB resident** Q4-class · compressed MLA, 16-bit cache · SSD tables: none · runtime: +8–16 GB Expanded-KV backends need ~95–105 GB cache instead.
Qwen3.8-Flash-Next Reasoning Released 26 Aug 2026	40	**~180B total · 6B active** MoE + lookup · 125B backbone	**~83 GB weights + ~4 GB cache ≈ 87 GB resident** MTPLX dynamic Q4 · 16-bit cache · +32 GB lookup tables on SSD · runtime: +4–8 GB
DeepSeek V4.1 Flash Max Released 10 Sep 2026	39	**~763B total · 16B active** MoE + Engram · 552B backbone · decode	**~316 GB weights + ~0.2–0.6 GB cache ≈ 317 GB resident** DwarfStar Q4 · packed FP4–BF16 cache scenarios · +203 GB Engram on SSD · runtime: +8–16 GB
Qwen3.8-27B Xhigh Released 14 Aug 2026	34	**27B total · 27B active** Dense	**20.4 GB weights + ~9.3 GB cache ≈ 30 GB resident** MTPLX dynamic Q4 · 16-bit cache · SSD tables: none · runtime: +4–8 GB
DeepSeek V4 Flash / 0731 Max · text model Released 31 Jul 2026 · checkpoint	34	**284B total · 13B active** MoE	**~155 GB weights + ~1 GB cache ≈ 156 GB resident** Q4_K_XL · compressed attention, BF16 cache · SSD tables: none · runtime: +8–16 GB
Claude Opus 4.6 BASELINE Max · AA estimate Released 5 Feb 2026	32	**Undisclosed** Architecture undisclosed	**Cloud only · frontier trajectory reference** AA marks 32 as estimated; independent evaluation forthcoming.
Qwen3.6-35B-A3B Reasoning · fallback Released 16 Apr 2026	18	**35B total · ~3B active** MoE	**~22 GB weights + ~3 GB cache ≈ 25 GB resident** Q4 · 16-bit cache · runtime: +4–8 GB